Describing a Distribution

AP Statistics · Topic 1.6 (CED 1.6) · B: 2026-09-17 · E: 2026-09-21 · TEACHER KEY

CED TETHER

- **Skill 2.A** — Describe data presented numerically or graphically.
- **Enduring Understanding UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.H** — Describe the characteristics of quantitative data distributions. [Skill 2.A]
- **EK UNC-1.H.1** — Descriptions of the distribution of quantitative data include shape, center, and variability (spread), as well as any unusual features such as outliers, gaps, clusters, or multiple peaks.
- **EK UNC-1.H.2** — Outliers for one-variable data are data points that are unusually small or large relative to the rest of the data.
- **EK UNC-1.H.3** — A distribution is skewed to the right (positive skew) if the right tail is longer than the left. A distribution is skewed to the left (negative skew) if the left tail is longer than the right. A distribution is symmetric if the left half is the mirror image of the right half.
- **EK UNC-1.H.4** — Univariate graphs with one main peak are known as unimodal. Graphs with two prominent peaks are bimodal. A graph where each bar height is approximately the same (no prominent peaks) is approximately uniform.
- **EK UNC-1.H.5** — A gap is a region of a distribution between two data values where there are no observed data.
- **EK UNC-1.H.6** — Clusters are concentrations of data usually separated by gaps.
- **EK UNC-1.H.7** — Descriptive statistics does not attribute properties of a data set to a larger population, but may provide the basis for conjectures for subsequent testing.

Essential question: What does a reader need from a description of a distribution before they can trust a summary statistic?

DOK-3 skill: 4.B · **FRQ pattern:** `two-descriptions-adjudicate-then-choose-center`

DOK rationale: Part (c) requires weighing two competing descriptions against specific graph features and defending a choice of summary statistic; no single procedure yields the answer, and either student could be defended on one feature alone.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Read the two descriptions aloud once; do not react to either.
- Start the video follow-along at minute 5. Collect nothing yet.
- Board slide back up at minute 33. Circulate with the printed prompts; confirm nothing.
- Collect sheets at the door. Score part (c) only, E/P/I.

Students do

- Write a one-sentence first take before anything is taught.
- Do the video follow-along (the DOK-1/2 work of the day).
- Finish (a)–(c) with the rules callout in front of them; complete the exit line; turn in.

Questions to ask

- Which feature of the graph did you look at first — why that one?
- If you deleted the 70-minute student, whose description changes?
- Ana says 'centered near 20'. Is she wrong about the center, or about something else?
- What would a reader who only saw Ana's sentence expect the longest commute to be?

Adult role
Solo teacher. Circulate 5 pairs. Answer questions with questions. Do not confirm a choice until the student has cited two features.

[WARN] WATCH FOR

- Reading 'skewed right' off the acronym without pointing at the tail.
- Treating the 70-minute student as typical, or ignoring the gap before it.
- Choosing the mean 'because it uses all the data' with no mention of the outlier.

SCORING PART (c) — E / P / I

Expected elements

- **picks-ben** (required) — Chooses Ben's description as the trustworthy one
- **cites-tail** (required) — Cites the right tail / asymmetry as a specific graph feature
- **cites-outlier** (required) — Cites the isolated 70–80 value (and/or the gap) as a second feature
- **median-with-mechanism** (required) — Reports the median AND explains that skew/outlier pulls the mean toward the tail

E	Picks Ben, cites the tail AND the outlier/gap, chooses the median with the pull-toward-the-tail mechanism stated in context.
P	Picks Ben with only one feature, OR chooses the median without the mechanism, OR gives the mechanism but names the mean.
I	Picks Ana, or chooses a measure of center with no reference to shape, or cites no specific graph feature.

Common mistakes

- Treats the 70-minute value as typical of the class
- Writes 'skewed right' from the acronym without pointing at the tail
- Says 'the mean is better because it uses all the data' without addressing the outlier
- Cites 'centered near 20' as wrong when the median really is in the 20s – the disagreement is about shape and the outlier, not the center

Optional #1 — not collected
DOK 2 Sketch a dotplot of 15 quiz scores (0–10) that is skewed LEFT, unimodal, and has one low outlier. Then write the one sentence a reader would need in order to describe it in context.

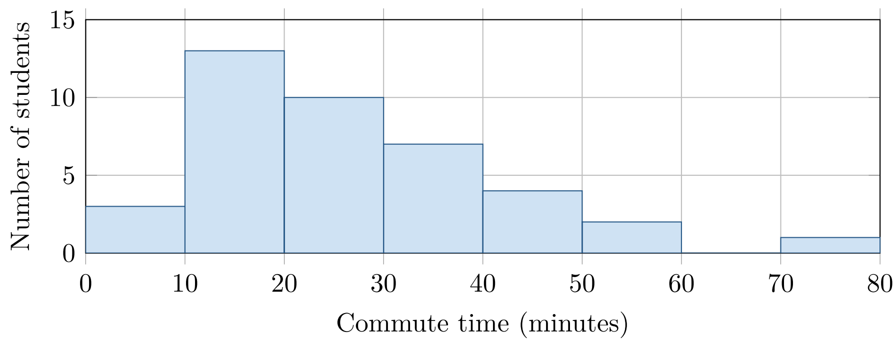
Key: Any dotplot with a cluster near 8–10, a thin tail to the left, and an isolated dot near 1–2. Sentence names the variable (quiz score), the shape, the center region, and the outlier.

[STAR] TODAY'S PROBLEM — annotated

The histogram shows how long it took 40 students to get to school one morning, in minutes. Two students described the distribution.

Ana: "Roughly symmetric, centered near 20 minutes, spread from 5 to 45 minutes."

Ben: "Skewed right with a peak in the 10–20 minute interval; most students take under 30 minutes; one student at about 70 minutes is an outlier."



FIRST TAKE (5 min)

Whose description do you trust more, Ana's or Ben's? One sentence. No wrong answers yet.
Key: Not graded. Expect a near-even split; the point is that they committed before the video.

Rules callout "DESCRIBING A DISTRIBUTION (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Name the shape of the histogram and state which interval contains the median commute time.

Key: Skewed right (unimodal). Cumulative counts 3, 16, 26: the 20th and 21st values fall in the 20–30 minute interval, so the median is in 20–30 min.

DOK 2 **(b)** Describe the distribution of commute times in context. Address shape, center, variability, and any unusual features, with units.

Key: Unimodal and skewed right. Center: median in the 20–30 min interval (mean about 27 min, pulled up by the tail). Variability: most commutes fall between 10 and 40 min; overall range about 5 to 80 min. Unusual: a gap in the 60–70 interval and one student in the 70–80 interval, an apparent high outlier. Context and units present.

DOK 3 **(c)** Whose description could a reader trust when choosing a summary statistic? Cite two specific features of the graph that settle it. Then state which measure of center you would report for these data and why.

Key: Ben. Two settling features: the long right tail (counts fall 13, 10, 7, 4, 2 to the right of the peak while there is only one bar to its left) and the isolated value in the 70–80 interval after a gap. Report the median: the distribution is skewed right with a high outlier, so the mean is pulled toward the tail and overstates a typical commute.

EXIT

One thing the video changed about my first take: ______________________________